Question 1

(a) (i) State **two** reasons why farmers would prefer to grow *Bt* corn. [2]

- (1) Increase crop yield or reduce loss of crop;
- As the toxin is targeted to kill the pests feeding on the crop;
- (2) Reduced spraying insecticide; cost-saving, maximise profits; reduce labour efforts;
- As the corn is genetically engineered to express own toxins;
- (3) Reduce the use of insecticide / accumulation of insecticide in environment;
- Preservation of natural environment / less harmful effects of insecticide on consumer;

(Max 1 for each reason)

Comments:
Producing Bt corn does not directly improve the quality of the crop. It serves more as a natural pesticide to target insects feeding on the corn. Hence improving yield here is really referring to decreasing loss. Note that Bt corn is not immune to pest attacking it.
Using pesticides on crops does not affect the quality of crops, however the potential side-effects on consumer's health is a valid concern and can be discussed as an advantage of Bt corn.
In order to reflect preference in the farmer's choice of Bt corn, students' answers should focus on reasons that would not be applicable to normal wild-type corn. Eg. '*Bt corn targets seed-corn beetles specifically and would not be harmful to humans'* would not exactly present a reason for preference since the wild-type corn would not cause any harm to humans when consumed as well.
The mechanism in which Bt toxin kills the insect is not relevant in this question.

(ii) With reference to Figure 1.1, explain how *Bt* gene from the bacteria is isolated and then inserted into corn cells. [3]

1. Both the *Bt* gene and plasmid treated with <u>same</u> restriction enzyme to
2. Generate restriction fragments with <u>complementary sticky ends</u>;
   (Accept blunt ends with use of terminal transferase)
3. Annealing via formation of hydrogen bonds / complementary base pairing occurs;
4. <u>DNA ligase</u> added to catalyse <u>formation of phosphodiester bonds</u>;
5. Recombinant plasmid <u>transformed</u> into vector *Agrobacterium tumefaciens*(ref. <u>heat shock & calcium chloride, electroporation, gene gun</u>);
6. Selection of transformed bacteria with recombinant plasmid (ref. selection markers, replica plating or blue white screening);

Comments:
Recombinant plasmids are first transformed into *Agrobacterium tumefaciens*, not to corn cells directly.
Growing Bt corn does not increases the quality of the crop.

(b) (i) With reference to Figure 1.2, compare the effect of consuming pollen from different corn plants on the survival of Monarch butterfly caterpillars. [3]

- Population of Monarch butterflies would decrease;
- Quote reference from question: 50% of Monarch butterflies found at maize growing areas OR only 55% of caterpillars survive 4 days after consuming Bt pollen OR Feeding period of caterpillars coincide with shedding of maize pollen between June and August;
- Disrupt ecological balance OR Affect biodiversity of environment;
- Due to decline in populations of predator that prey on Monarch butterflies OR Due to reduced pollination by Monarch butterflies of the surrounding flowering plants;

Comments:
Reject students discussing the evolution of the organisms such as Monarch butterflies or the milkweed plants.
Question is asking for the effect on the **environment**. Many students focused only on the impact on the population of Monarch caterpillars or the Monarch species and how the allele frequency for resistance would increase in the population etc. The complete answer has to link back to influences on environment, and not on evolution.
Reject possibility of 'superweeds' as the question asked for use of information **provided in the question only**, which revolved around the survivability of Monarch butterflies. Hence the answer should focus on impacts on the environment due to effect (decline) on the Monarch butterflies population because of caterpillars ingesting Bt toxin.

(ii) Using only information given in this question, discuss the effect of continued cultivation of *Bt* corn in the USA on the environment in the corn-growing area. [2]

- Conduct thorough checks / testing; before released / approval for sale for consumption;
- Relevant government bodies to enforce strict regulation / ban on import of GM food that failed the tests;
- Clear labelling of GM food; such that consumers can be aware of its content or able to make their own choice;
- Education of public of the risks and benefits of GM food;

Comments:
Reject: Consumers must be discerning towards GM and non-GM food. Consumers must make a wise choice / consumers must check the label properly … etc. That is any answers that focused on the consumers taking action.
The question is on how consumers can be **protected**. Hence answers are expected to describe efforts from external factors / organizations to reduce the concerns related to genetic engineering and not from the individual perspective of the consumers.

(c) (i) Suggest how consumers can be protected from these concerns that arise. [2]

- Caterpillars that consumed pollen from genetically modified corn (group A)showed decrease in % survivability;
- From 100% to 55% by day 4;
- Steady, consistent decline per day;
- Ref. **effect of Bt toxin** coded by Bt gene causing death of caterpillars;
- Caterpillars that consumed pollen from non-genetically modified corn (group B) **remained unaffected from pollen**;
- Survival remained at 100% throughout 4 days;
- Likewise for caterpillars that consumed leaves with no pollen (group C):no effect / remained unaffected;
- As it is acting as control to show effect due to consumption of pollen;

Comments:
Reject: % Survival remained unchanged / constant for group B and C without stating the effect.
The question is asked on the effect of ingesting various pollen. Most students describe and quote the figures well, but stopped short at *stating and comparing the effect* directly. For group B (and C) where no effect is observed, students do have to identify and state it as a form of comparison to group A where survivability decreased.

<u>Question 2</u>

(a) (i) Describe how explants may be grown into callus. [3]

- explants are transferred **aseptically**/ reference to **aspetic**/ **sterile condition** to prevent microorganisms growth/ use of laminar flow cabinet;
- to agar plate/ culture vessels / **culture medium**;
- containing nutrients such as a **sucrose as a carbon source**/ source of energy/ respiratory sunstrate**,**
- **organic nutrients**/ amino acids and vitamins **and inorganic ions**/ minerals/ named 2 e.g. nitrates, potassium, phosphates, etc.
- and **plant growth regulators**;
- of **intermediate ratio of auxin and cytokinin**;
- to stimulate growth to form callus tissue **by mitosis**;

Comments:
Rejected mere mention of surface sterilization of explants as an indication of use of aspetic condition because this is one of the sampling method used which actually resulted in high contamination.

explant	time of year	number of explants	number of cultured explants with **no** fungal or bacterial contamination	Percentage of cultured explants with **no** fungal or bacterial contamination/ **%**
leaf disc	April	153	12	**8**
	August	322	16	**5**
	January	332	30	**9**
stem tissue	April	194	116	**60**
	August	191	122	**64**
	January	211	156	**74**

Table 2.1

(ii) With reference to **Table 2.1**, compare the effect of the different methods of taking tissue samples and the time of year on fungal and bacterial contamination of the cultured explants. [3]

- **stem tissue has less contamination** than leaf discs for all three times of the year;
- for example, in April, 60% of stem tissue has no fungal or bacterial contamination compared to only 8% of leaf discs or other appropriate comparison with manipulated data;;
- both treatments have **highest % non-contamination in January** of 74% for stem tissue and 9% for leaf discs respectively;;
- lowest for leaf discs in August of 5%, for stem tissue in April of 8%;;

Comments:
Students need to take note that there is a need to manipulate the data to show % as the number of explants taken were different so the low numbers with no contamination may be simply due to less explants taken.

(iii) Suggest a possible reason for the differences observed between the two types of explant. [2]

- leaf discs are **exposed to the external environment** and **bleach may not** be able to **remove all the contaminants** on them;;

- while using sterile hypodermic syringe to obtain the stem cells **directly** remove the sample hence the danger of **exposure to contaminants** in the external environment is **minimised**;;

Comments:
Major misconception comes in the form of stem tissue being obtained from meristemic regions of the plant (confusion with the term stem cell in animal?) but there is no indication that this is obtained from the shoot tip.

(b) (i) Explain the purpose of calli **A** and **B** in this investigation. [2]

A:
- Calli A grew minimally / grew the least at all concentrations of herbicide tested;
- to show that normal cells have minimal resistance/ do not have resistance to the herbicide;
- to serve as a base line for comparison of **C and D**;

B:
- Calli B shows that the normal ALS gene from weed cannot cause herbicide resistance/ genetic engineering technique itself does not cause resistance (c.f. to A)
- so the results in **C and D** are due to the presence of mutation in gene from weed;;

Comments: Students should note that A and B are the controls in the investigation and that the size of callus of 1 mm is an indication that the herbicide has cause the callus not to grow at all, i.e. there is no resistance to herbicide at all. Some students merely mention that they are acting as control to show the effect of natural growth of calli for A and the effect of growth of calli in the presence of normal ALS gene for B are not giving a reason for using them at all for the purpose is still unclear.

(ii) With reference to **Figure 2.1**, assess the effect of the mutated ALS gene on the expression of sulfonylurea resistance. Justify your answer. [3]

1) Presence of mutated ABL gene causes herbicide resistance (regardless of no. of copies) at low concentration of herbicide;

2) The conconcentration of herbicide that the cells can withstand is dependent / increases with the number of copies of the mutated ALS gene present/ reference to more copies of mutated ALS gene, greater resistance;

3) D has two copies of the mutated ALS gene is resistant even up to the highest conc. of herbicide tested / 1.00mg/dm^{-3};

4) average diameter of calli of D remain at 5 mm/ grow to maximum diameter of 5 mm at all herbicide concentrations tested

5) C has only one copy of the mutated ALS gene is only completely resistant to the herbicide up to concentrations of 0.05 mg dm^{-3}

6) After which the higher the herbicide conc. the greater the inhibition on callus growth of C as the size decreases from 5 mm to 3 mm at 0.4 mg dm^{-3};

(Points 5 and 6 mark for partially resistant to herbicide for one copy of mutated gene and QV)

7) Calli for C & D are larger than that for controls (A & B) at all concentrations of herbicide tested except at 1.00mg/dm^{-3} where C has a callus of the same average diameter as controls/ reference to comparison with controls to show that mutated ALS gene is responsible for sulfonylurea resistance;

8) Above 0.2 mg dm^{-3}, average diameter is 1 mm for A and B which is the size of the callus due to no resistance to herbicide;

Comments:

1. Students need to take note that this is GE technique, there is no heterozygous for 1 copy of mutated gene and homozygous when referring to 2 copies of mutated gene because rice cells in the first place do not have any copies of the mutated gene until they are treated.

2. Some students explained for why there is difference in the results in C and D using underlying principles glimpsed from the passage given but there is no need to do so as justifying your answer merely requires you to give support from the graph after you have ascertained that there is an effect of the mutated gene on resistance.

Question 3

(a) Explain how CF is inherited. [2]

- autosomal recessive;
- *autosomal* –males and females are equally affected/ expression of characteristic not dependent on the sex of the individual;
- *recessive* – requires both alleles to be defective before the child is phenotypically CF; (®: individual must be homozygous recessive becos recessive is not explained)
- carrier x carrier – ¼ chance of producing a CF child;

(b) Using the information provided in Figure 3.1, assess the suitability of DM1 and DM2 as DNA markers. [3]

- DM1 – more suitable /DM2 – less suitable;
  (mark awarded only if the student is able to provide some support for the answer.)

  Either
- Becos for DM1, 7 out of 8 (88%) of the CF individuals have the **only one** **5kb band**/ **homozygous for the 5 kb fragments;;**
- 5 out of 8 (62.5%) of the CF individuals have **only one** **4kb band**/ **homozygous for the 4 kb fragments** for DM2;;

  Or
- For DM1 - All CF individuals have **only one** **5kb band**/ **homozygous for the 5 kb fragments** except II-3 who has 2 bands 5kb and 7 kb**;;**
- For DM2 - 5 CF individuals have **only one** **4kb band**/ **homozygous for the 4 kb fragments** and 4 CF individuals have 2 bands 3kb and 4kb / about equal number of CF individuals and normal individuals have a single 4kb band as well as 2 bands 3kb and 4kb**;;**
  (So the inheritance of the DM1/5kb fragment is more consistent compared to DM2)

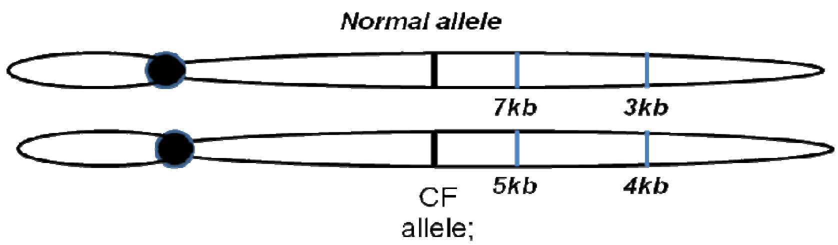
Comments:
There are many ways of approaching this question. Students should note that merely quoting who is showing the 'right' banding pattern and who is not cannot be considered to be "*assessing* the suitability" of molecular markers. Students should attempt to sum up the observations/data given eg. 7 out of 8 CF individuals when considering the entire family or 6 out of 7 CF individuals when considering the children only.
Molecular markers are non-coding and therefore it is not appropriate to use the term homozygous recessive when describing the presence of two 5kb fragments.

(c) Using your understanding of linked genes, provide a possible explanation for your answer to (b) above. [2]

- DM1 **closer** to CF gene compared to DM2;;
  (note: some form of comparison required between DM1 and DM2 before a full mark is given.)
- Higher tendency for the 5kb fragment to be inherited together with the CF gene;
- Hence lower number of recombinants (individuals II-3 and II-10 compared to DM2 (II-1,2,,6,7 and 10) ;

Comments:
No mark is given if students do not answer in context of DM1 and DM2. Question specifies that an explanation is required in terms of why DM1 is more suitable as a DNA marker based on your understanding of linked genes.

(d) One possible arrangement of the DM1and DM2 alleles wrt to the CF gene for Individual II-4.



Mark scheme: (i) correct allele for CF;
(ii) correct alleles for DM1; and DM2;
(iii) distance between DM1 and CF is <u>shorter</u> than DM2 and CF;

(e) Describe one ethical issue that can arise from such diagnosis. [2]
Either
- If a fetus is diagnosed to have a high chance of being afflicted with CF;;
- Would be parents may resort to **abortion** because of concerns about the quality of life for the child & medical costs incurred;;

Or
- Risk profiles of fetuses known to insurers/Fairness in the use of genetic information by insurers;;
- Insurers may reduce coverage or increase insurance amount;;

(f) (i) Explain why there seems to be fewer objections to this method compared to one where gene therapy is applied to gametic cells.

Somatic cell gene therapy: (1 max)
- Only the individual receiving the treatment is affected;
- The allele that is introduced is not inherited by his offspring;
- Individual has consented to the therapy;

Gametic cell gene therapy: (1 max)
- Inheritable by the offspring/ future generations;
- Such treatment produces generations of unconsenting research subjects;
- Long term effects of gene therapy still unknown;
- <u>*May*</u> tamper with human evolution eg. genes which cause harm in one environment may be of evolutionary significance in another;

(ii) Explain why gene therapy provides a possible cure but not stem cell therapy. [2]
- Multiple organs affected – lungs, pancreas etc/ many different cell types are involved;;
- and not all organs involved have stem cells that can be used for stem cell therapy/ difficulty in extracting the stem cells for use;;Difficulty in obtaining stem cells that are able to differentiate into many different types of cells;;

Question 4

INTRODUCTION/Theoretical Background (2 marks)

1. Explanation of theory to support practical procedure
- Define transformation as uptake of foreign DNA (pGLO plasmid) from the external environment by *E. coli* which expresses the genetic information encoded by the plasmid.
- antibiotic resistance gene on plasmid will confer ampicillin resistance to the bacterial cells → Able to survive when exposed to ampicillin to form colonies,
- arabinose induce expression of GFP gene → green colonies under UV light

2. Scientific reasoning behind the method
- Ca^{2+} ions in transformation solution neutralize the negative charges of the plasmid DNA to facilitate its entry into the bacteria
- Heat shock methods create temporary pores in bacterial cell membrane to facilitate the entry of plasmid

PROCEDURE/METHOD (Total number of points = 10 [3 – 9 & 11]; max is 6 marks)

3. Preparation of different nutrient agar plates
(table/ any equivalent to show labels and composition of each plate)

4. Chemical-heat shock method for transformation
a) competent cells + plasmid solution / sterile water+ transformation solution in a tube)
b) ice → 42°C, 50s heat shock → ice again
c) Method of achieving and maintaining desired temperature

5. **a)** Method of plating
Removal of suitable amount of bacteria → appropriate plate → spreader/ inoculating loop/ beads (spreading)

b) Incubation of bacteria for growth
(incubator oven, 37°C, overnight)

6. Use of control plate (any one)
(What – Grow the bacterial cells treated with sterile water instead of pGLO plasmid solution on LB agar plates
Why – to show that bacterial cells are still alive after going through the transformation procedure/ to compare with those cells grown on LB/Amp agar plates to show that ampicillin in the agar plate is effective is killing bacteria cells with no resistance
What – grow bacterial cells treated with sterile water on LB/Amp agar plates
Why – To compare with pGLO treated cells grown on LB/Amp agar plates to show that ampicillin resistance is indeed conferred by the pGLO plasmid
What – Grow pGLO treated bacterial cells on LB/Amp agar plate
Why – To compare with LB/Amp/Arabinose agar plate to show that Arabinose is causing fluorescence in pGLO treated bacterial cells)

7. Use of sterile equipment to prevent cross contamination between plates
(Use different sterile glass beads/ spreader/ inoculating loop for spreading the cells evenly across plate and new micropipette tips for every transfer of volume)
No need to have all, just the indication that of what the student will do to prevent cross contamination.

8. Repeat twice with replicates
(Idea of consistency → 6 sets of data)

9. Method of determining successful transformation
(White colonies / Green colonies and **count** the number of colonies that grow on nutrient plates)

USE OF DATA TO REACH A CONCLUSION

10. Labeled table for data collection **(1 mark)**
(Independent variable in leftmost column, headings: Name and type of agar plate, Observations: description of bacterial growth & (average) number of colonies; Experiment and replicates ® R without legend)

11. Processing of data
(Calculate **average** number of bacterial colonies, Formula)

12. Graph to represent processed data **(1 mark)**
(bar chart: Plot **average** number of colonies against Name and type of agar plate – labels for x-axis, y-axis and shape of graph?)

SAFETY ISSUE

13. Safety in handling of bacteria **(1 mark)**
(e.g. heat sterilization of bacterial plates at 121°C/ autoclaving before disposal / wear gloves when handling bacteria)
There must be a What, Why and How.

14. The use of correct technical and scientific terms **(1 mark)**

Essay Answer

5 **(a)** Compare and contrast the use of viral and non-viral mode of delivery in gene therapy. [6]

	Viral	Non-viral
1. Example	Adenovirus, retrovirus	Liposomes
2. Size	Smaller gene insert (8 – 20kb)	Larger gene insert (>20kb)
3. Immune response	More likely to evoke immune response	Less likely to evoke immune response
4. Disease causing / risk assessment	Potential to cause disease / more dangerous	Do not cause disease at all / safer
5. Specificity	More specific targeting of cells	Less specific targeting of cells
6. Integration	Possible to integrate into genome (using Retrovirus) / cause insertional mutagenesis	Not able to integrate into genome (extrachromosomal) / do not cause insertional mutagenesis
7. Entry into cell	Viral infection (fusion and endocytosis)	Fusion of lipid bilayer with cell membrane
S1	Both involve introducing a **functional allele** into cells;	
S2	Both can be used ex vivo and in vivo;	
S3	Accept max 1 similarity in limitation of gene therapy;	

Comments:
Some students included comparisons between viruses in viral approach (eg. the need for rapidly dividing cells for retrovirus) rather than between viral and non-viral approach.
Note that there is no clear distinction in gene expression based from the two delivery systems. Expression is transient in liposomes and adenovirus, and highly variable in retrovirus.

Students are misusing the terms ‘allele’ and ‘gene’ in this context, where the copy of **functional allele** is introduced through gene therapy, not gene.

5(b) Explain how probes can be used to detect and analyze restriction fragment length polymorphism (RFLP). [8]

What is a probe/ Characteristics of a probe

1. A single-stranded DNA/ made up of nucleic acids;
2. Probe is complementary to ssDNA (target sequence);
3. Binds to target sequence via hydrogen bonding;
4. probe is radioactively labeled;
5. Presence can be detected by autoradiography;

How to detect RFLP: Nucleic acd hybridisation

6. In RFLP analysis, DNA is first digested by a restriction enzyme;
7. Digest run on a Gel electrophoresis to separate DNA fragments by size;
8. Gel exposed to salt and alkaline solution to denature double stranded DNA into singled stranded DNA;
9. Transfer to nitrocellulose filter by capillary action;
10. Filter baked at 80oC to permanently fix the single stranded DNA;
11. Introduce probe to allow for hybridization (of probe to specific ssDNA at the region complementary to the sequence of probe);
12. Wash off excess probe;
13. Subject nitrocellulose membrane to autoradiography/ put an X-ray film onto the filter;
14. Where the radioactive probe binds to the target sequence, a **dark image** will be produced on the X-ray film; (This is the detection part as it allows visualization of the target sequence.)

(6 max)

Analysis of RFLP

15. Disease detection: (2m max)
    a) If a mutation causes a **loss/gain of restriction site** in the disease-causing allele;
    b) There will be **different restriction fragment lengths obtained**;
    c) If probe binds to a common sequence in both the alleles, different number of bands or different sized bands will be obtained;
    d) Detection of a specific RFLP fragment by a probe (that is linked to the disease-causing allele) will allow the detection of the disease-causing allele i.e. determine by proxy as the RFLP and disease-causing allele tend to be **inherited together;**

16. Genomic mapping (1m)
    a) Probe can be used to follow the inheritance of **non-coding DNA** that are linked;
    b) Allows the detection of cross overs/ Use COV/ equation/ proportion of recombinants;
    c) to determine **relative distance** between the two RFLP;
    d) The higher the number of recombinants, the higher the COV, the further the 2 RFLPs are;

17. Genetic Fingerprinting (1 m)
    a) Probe binds to a common sequence that is outside the VNTR;
    b) Allows the detection of VNTR of different sizes/ show up as different banding patterns for individuals;

5(c) Discuss the benefits of the human genome project. [6]

M	**Molecular Medicine**
1	Improved (prenatal) diagnosis of disease / Early detection of genetic predispositions to disease
2	Design drugs and control systems for drugs based on molecular information
3	Design custom drugs based on individual genetic profiles for greater efficacy
4	Design molecular treatment such as gene therapy
R	**Risk assessment**
1	Assess risks of individuals for certain diseases
2	To assess the difference in responses to chemicals in individuals due to genetic variation
F	**DNA forensics**
1	Identify endangered and protected species as an aid to wildlife officials
2	Identification of individuals through matching of DNA left at crime scenes
3	Establish paternity and other family relationships
E	**Human evolution and comparative genomics**
1	Understand human evolution such as migration of humans
2	Comparing genomics between human and other organisms to uncover functions of genes (in diseases and other traits)
*	Energy and environmental applications in microbial genomics
*	Agriculture, livestock breeding and bioprocessing